

Introduction to L^AT_EX

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1 Document structure

L^AT_EXdocument can be divided into three main parts;

1. Document class
2. Preamble
3. Document body

1.1 Document class

This is the first line of L^AT_EXdocument, and defines the type (or class) of document. Some of the common classes include:

- `article` → to write articles/short reports. Top level header is `section` and from there it goes to `subsection`, `subsubsection` and so on.
- `book` → for books. Supports `part` and `chapter` levels for writing. And if I recall correctly, has different margins for odd/even pages.
- `beamer` → for creating presentations.

In addition to defining the type of document, the command `\documentclass` also supports optional arguments. Some of the options are:

- `papersize` → a4, a5, letter etc.
- `fontsize` → font size are defined in terms of *pt*. Default is 10pt.
- `orientation` → landscape/portrait. Default is portrait.

An example of document class is:

Listing 1: Document class

```
1 \documentclass[a4paper, 10pt]{article}
```

1.2 Preamble

This section defines all the packages that are required for the document, settings that apply to whole document, user based commands (or macros) as well as document metadata. An example of preamble is presented in Listing 2:

Listing 2: Defining document preamble

```
1 % packages
2 \usepackage[colorlinks=true]{hyperref}
3 \usepackage[utf8]{inputenc}
4 \usepackage[T1]{fontenc}
5 \usepackage[left=2.54cm, right=2.54cm, top=2.54cm,
6   bottom=2.54cm]{geometry}
7 % settings for document
8 \setlength{\parindent}{0pt} % set indentation from left at beginning
9   of paragraph.
10 \setcounter{tocdepth}{2} % table of contents depth
11 % metadata
12 \author{names}
13 \title{text}
14 \date{text}
```

1.3 Document body

This is the part where actual text (text that you want to put in document) goes. The contents are enclosed by `\begin{document}` and `\end{document}` commands. Inside document body, one can include different environments like figures, tables, equations and so on.

Some common formatting options:

- `\textbf{Bold text}` → **Bold text**
- `\textit{Italic text}` → *Italic text*
- `\underline{Underline text}` → Underline text
- `A\textsubscript{subscript}` → $A_{\text{subscript}}$
- `A\textsuperscript{superscript}` → $A^{\text{superscript}}$
- `\textcolor{Blue}{blue text}` → blue text
- `\newline` or `\\` → for line break.
- `\newpage` → forcing page break.

A simple example for document body is presented in listing 3. The command `\maketitle` generates a title from metadata as defined in Section 1.2. The command `\tableofcontents` produces table of contents (toc). By default toc depth is 3 for the article class, meaning headings upto and including `\subsubsection{}` are included in the toc. This can be modified using `\setcounter{tocdepth}{2}` in document preamble. In this particular example the depth is set to 2, meaning only `\section{}` and `\subsection{}` are included in toc.

Listing 3: Example for document body

```
1 \begin{document}
2
3   \maketitle
4   \tableofcontents
5   \newpage
6
7   \section{Introduction}
8
9   \section{Material and methods}
10
11  \subsection{Chemicals}
12
13  \subsection{Analytical methods}
14
15 \end{document}
```

1.3.1 Section commands

In L^AT_EX the document can be structured into chapters, section, subsection and so on. The nesting order is:

1. `\part{}`
2. `\chapter{}`
3. `\section{}`

4. `\subsection{}`
5. `\subsubsection{}`
6. `\paragraph{}`
7. `\subparagraph{}`

In `\documentclass{article}` the top level heading is `\section{}` and `\part{}` or `\chapter{}` throws an error. Those levels are available in book or report class.

In addition, there are also commands with asterik (*) at their end. This suppresses the numbering and exclude the header from table of contents. See

2 Equation

Writing equation in L^AT_EX is very simple. Equations can be written in line using \$ operator, or they can be written inside math environment. See listing 4 for example

Listing 4: Writing equations in L^AT_EX

```

1 Equations can be written inline e.g. Relativity states $E=mc^{2}$
2
3 Equation can also be written in display mode using the
  \texttt{equation} environment:
4 \begin{equation}
5   E = mc^{2}
6 \end{equation}
```

Equations can be written inline e.g. Relativity states $E = mc^2$

Equation can also be written in display mode using the `equation` environment:

$$E = mc^2 \tag{1}$$

Spacing is ignored in math mode. To add space between characters or operators use backslash (\) followed by space.

2.1 Sub-equations

L^AT_EX also supports writing sub-equations (listing 5). Writing math characters¹ is also quite intuitive, normally the character is preceded by a backslash (\) e.g. `$\rho$` $\rightarrow \rho$. Note that for writing Greek/Latin characters within text requires the same syntax as inline equations, i.e. enclosing it between \$ operator. Few more things to note in listing 5;

- the command `\label{key}` is used to create an anchor for cross-referencing the object at some other point in document.
- Math (or equation environment) normally writes everything in *italics*. `$\mathrm{}` formats the math symbols to normal (roman) font.
- Simple operators like addition and subtraction can be written as keyboard characters (+ or -). For multiplication use `\times` or `\cdot` $\rightarrow \times$ or $\cdot$ respectively. For division, use `\frac{num}{den}` $\rightarrow \frac{num}{den}$

¹For a full list of L^AT_EX symbols see [The Comprehensive L^AT_EX Symbol List](#)

Listing 5: Writing subequations

```

1 \begin{subequations}
2 \begin{gather}
3 \mathrm{uc}_f|_{z=0} = \mathrm{uc}_f|_{z=0^+} - D_{ax} \frac{\partial c_f}{\partial z} \Big|_{z=0^+} \quad \text{\label{eq:BC-inlet}} \\
4 \mathrm{frac{\partial c_f(z=L,t)}{\partial z} = 0 \quad \text{\label{eq:BC-outlet}} \\
5 \end{gather}
6 \label{eq:BoundaryConditions}
7 \end{subequations}

```

$$uc_f|_{z=0} = uc_f|_{z=0^+} - D_{ax} \frac{\partial c_f}{\partial z} \Big|_{z=0^+} \quad (2a)$$

$$\frac{\partial c_f(z=L,t)}{\partial z} = 0 \quad (2b)$$

2.2 Multiline equations

Sometimes the equations are too long to be properly displayed on a single line. In that case use `split` or `multiline` environment to write multiline equations with predefined line breaks. See listing 6 for code, and eqs. (3) and (4) for 'output.

Listing 6: Writing multiline equations in L^AT_EX

```

1 \begin{multiline}
2 \mathrm{frac{\partial c}{\partial t} = -u \cdot \frac{\partial c_{i+1,j}}{\partial x} - v \cdot \frac{\partial c_{i,j+1}}{\partial y} - } \\
3 \mathrm{+ \mathbb{D} \left( \frac{c_{i+1,j} - 2c_{i,j} + c_{i-1,j}}{\Delta x^2} + \frac{c_{i,j+1} - 2c_{i,j} + c_{i,j-1}}{\Delta y^2} \right) + R} \\
4 \label{eq:multiline}
5 \end{multiline}
6
7 \begin{equation}
8 \begin{split}
9 \mathrm{p(\theta|y,n)} \propto \mathrm{propto} \theta^y (1-\theta)^{n-y} \theta^{\alpha-1} (1-\theta)^{\beta-1} \\
10 \mathrm{\propto} \theta^{y+\alpha-1} (1-\theta)^{n-y+\beta-1} \\
11 \mathrm{\propto} \mathrm{Beta}(\alpha+y, \beta+n-y) \\
12 \label{eq:align}
13 \end{split}
14 \end{equation}

```

$$\frac{\partial c}{\partial t} = -u \cdot \frac{c_{i+1,j} - c_{i-1,j}}{2\Delta x} - v \cdot \frac{c_{i,j+1} - c_{i,j-1}}{2\Delta y} + \mathbb{D} \left(\frac{c_{i+1,j} - 2c_{i,j} + c_{i-1,j}}{\Delta x^2} + \frac{c_{i,j+1} - 2c_{i,j} + c_{i,j-1}}{\Delta y^2} \right) + R \quad (3)$$

$$\begin{aligned} p(\theta|y,n) &\propto \theta^y (1-\theta)^{n-y} \theta^{\alpha-1} (1-\theta)^{\beta-1} \\ &\propto \theta^{y+\alpha-1} (1-\theta)^{n-y+\beta-1} \\ &\propto \mathrm{Beta}(\alpha+y, \beta+n-y) \end{aligned} \quad (4)$$

3 Figures

Figures in $\text{\LaTeX}$ are included using `figure` environment. The supported types are `.pdf`, `.eps`, `.png`, `.jpeg`. General syntax for including figures is given in listing 7. Important to include packages `graphicx` and `float` in document preamble. The latter is required for changing figure position.

Listing 7: Including figures in $\text{\LaTeX} 2_{\epsilon}$

```
1 \begin{figure}[H]
2 \centering
3 \includegraphics[width=0.6\textwidth]{learning_curve.png}
4 \caption[Comparison between different tools for writing
   text]{Comparison between different tools for writing text. \LaTeX{}
   is a bit difficult to start with, but becomes easier to write with
   time. Markdown is also a nice way of writing documents. See
   \href{https://quarto.org/docs/authoring/markdown-basics.html}{Quarto}
   for a good markdown option.}
5 \label{fig:demo_figure}
6 \end{figure}
```

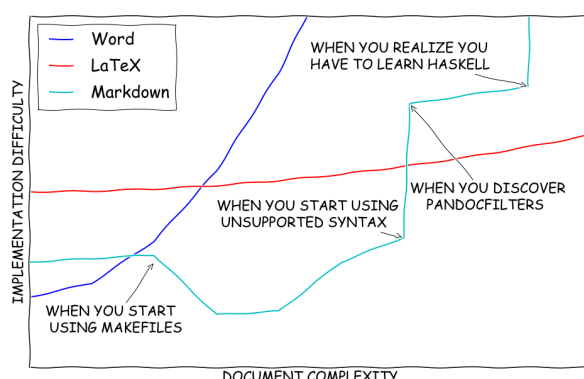


Figure 1: Comparison between different tools for writing text. $\text{\LaTeX}$ is a bit difficult to start with, but becomes easier to write with time. Markdown is also a nice way of writing documents. See [Quarto](https://quarto.org/docs/authoring/markdown-basics.html) for a good markdown option.

Some notes for listing 7;

- `[H]` right after figure environment tells the compiler to place figure precisely at that point. Other options include `t`; page top, `b`; page bottom, `h`; approximately here.
- `\centering` is used to instruct the compile to horizontally center figure.
- `\includegraphics` is the command that actually includes the figure. The argument in `[]` are optional. There are several options including some for trimming the figure, but normally I only use sizing (`width`) argument. In my experience, cropping the figure or removing white spaces in figures is more easy using external tools like adobe or inkscape.
- `caption` command has two arguments:
 - `[short caption]` - optional argument that appears in list of figures if included in document. List of figures can be generated using command `\listoffigures` inside document body.

– `{long caption}` - required argument, which appears below the figure.

- `\label{key}` to create anchor for cross-referencing.

3.1 Subfigures

Often one wants to include multiple figures in a panel layout. This can be easily done using the `subfigure` environment which requires loading the `subcaption` package in preamble. See listing 8 for source code for Figure 2. Captions and labels can be provided individually for each subfigure. e.g. `\Cref{fig:ReLU-Function}` → Figure 2a.

In listing 8, each subfigure is assigned a fixed width upfront e.g. `\begin{subfigure}{0.49\textwidth}` tells the compiler to use 49% of available textwidth for that figure. One can also use `\linewidth`, but this may cause the figure to extend beyond normal text margins.

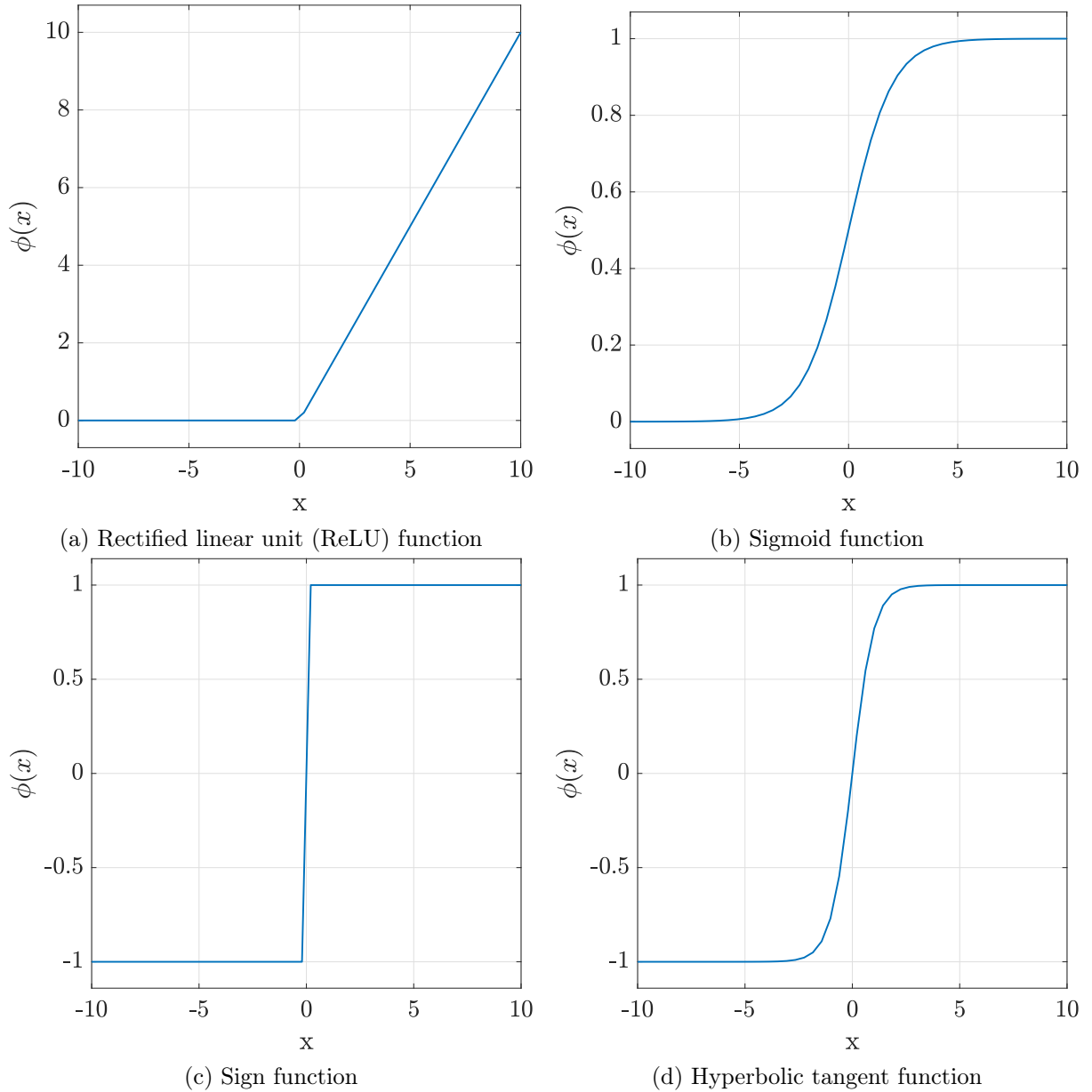


Figure 2: Some commonly used activation functions for neural networks

Listing 8: Creating subfigures in L^AT_EX

```

1 \begin{figure}[H]
2   \centering
3   \begin{subfigure}{0.49\textwidth}
4     \centering
5     \includegraphics[width=\textwidth]{ReLU.pdf}
6     \caption{Rectified linear unit (ReLU) function}
7     \label{fig:ReLU-Function}
8   \end{subfigure}
9   \hfill
10  \begin{subfigure}{0.49\textwidth}
11    \centering
12    \includegraphics[width=\textwidth]{SIGN.pdf}
13    \caption{Sign function}
14    \label{fig:Sign-Function}
15  \end{subfigure}
16
17  \begin{subfigure}{0.49\textwidth}
18    \centering
19    \includegraphics[width=\textwidth]{SIGMOID.pdf}
20    \caption{Sigmoid function}
21    \label{fig:Sigmoid-Function}
22  \end{subfigure}
23  \hfill
24  \begin{subfigure}{0.49\textwidth}
25    \centering
26    \includegraphics[width=\textwidth]{TANH.pdf}
27    \caption{Hyperbolic tangent function}
28    \label{fig:Tanh-Function}
29  \end{subfigure}
30  \caption{Some commonly used activation functions for neural networks}
31  \label{fig:Neural-Network-Activation-Functions}
32 \end{figure}

```

3.2 Caption alignment

By default captions are centered in L^AT_EX. However one can change them to left or right using the commands listed in listing 9 (Figure 3). This requires loading `caption` package to be loaded in preamble. For right alignment one can replace `raggedright` with `raggedleft`. Notice that the `\captionsetup{}` is specified within the `figure` environment, so only the caption for this figure will be affected. To avoid repeating this for every figure, use the same command in document preamble after loading the `caption` package.

Listing 9: Figure caption alignment

```

1 \begin{figure}[H]
2   \captionsetup{justification=raggedright,singlelinecheck=false}
3   \centering
4   \includegraphics[width=0.6\textwidth]{ReLU.pdf}
5   \caption{Left aligned captions}
6   \label{fig:Left-caption}
7 \end{figure}

```

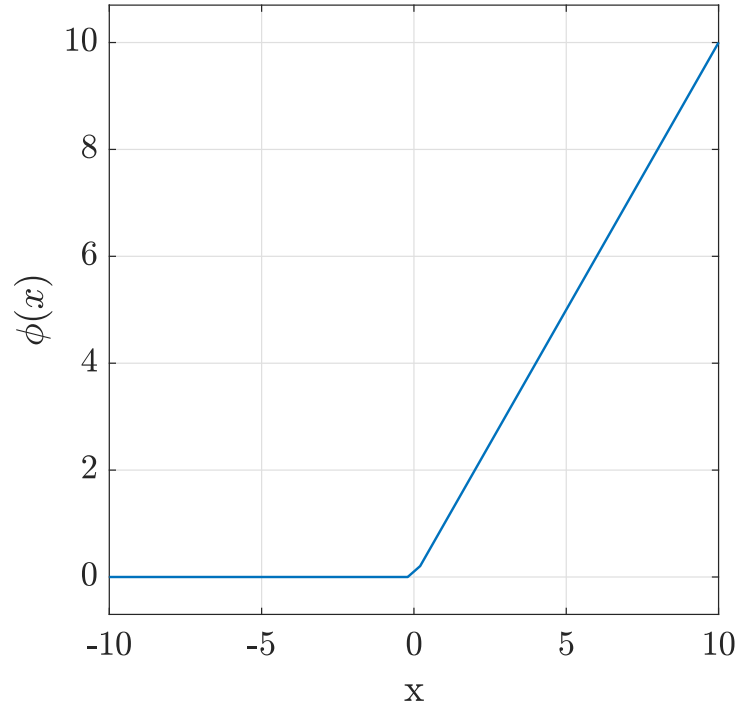



Figure 3: Left aligned captions

4 Tables

Default tables in $\text{\LaTeX}$ don't require any package. However, they are limited in their output. For this reason we use the `tabulararray` package, that provides support for long tables as well as other useful functionalities. An example of long table generated using `tabulararray` is provided in

Listing 10: Using `longtblr` from `tabulararray` for tables

```

1 \begin{longtblr}
2   [
3     caption={Commonly used activation functions for neural networks},
4     label={tab:ActivationFcns}
5   ]
6   {
7     colspec={|[2pt]X[4cm,m,j]|X[3.5cm,m,c]|X[4cm,m,c]| [2pt]},
8     rowhead={1},
9     row{1} = gray4,
10    hline{1,Z}={2pt},
11    hline{2-Y}={1pt}
12  }
13  \textcolor{white}{\textbf{Activation Function}} &
14    \textcolor{white}{\textbf{Equation}} &
15    \textcolor{white}{\textbf{Graphical representation}} \\
16  Sigmoid function & \SetCell{mode=dmath} \phi(x) = \frac{1}{1+e^{-x}} & \\
17    & \begin{minipage}{\linewidth}
18      \includegraphics[width=\textwidth]{Figures/SIGMOID}
19    \end{minipage} \\
20  Hyperbolic tangent function & \SetCell{mode=dmath}
21    \phi(x)=\frac{2}{1+e^{-2x}}-1 & \begin{minipage}{\linewidth}
22      \includegraphics[width=\textwidth]{Figures/TANH}
23    \end{minipage} \\
24  \end{longtblr}

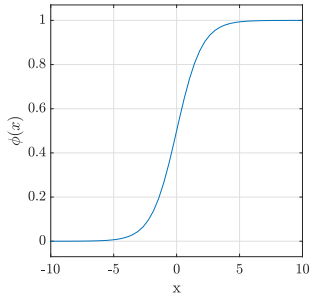
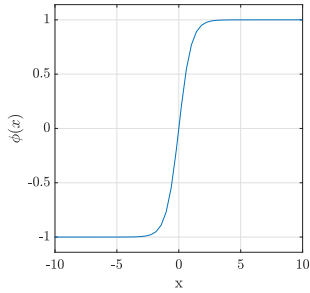
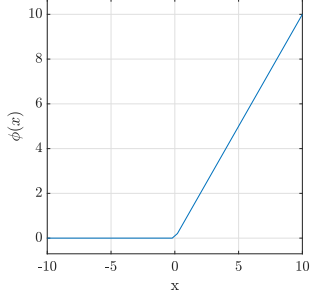
```

```

20 Rectified linear unit (ReLU) function & \SetCell{mode=dmath} \phi(x)
    = \max(0,x) & \begin{minipage}{\linewidth}
21 \includegraphics[width=\textwidth]{Figures/ReLU}
22 \end{minipage} \\
23 Sign function & \SetCell{mode=dmath} \phi(x) = \text{sign}(x) &
    \begin{minipage}{\linewidth}
24 \includegraphics[width=\textwidth]{Figures/SIGN}
25 \end{minipage}
26 \end{longtblr}

```

Table 1: Commonly used activation functions for neural networks

Activation Function	Equation	Graphical representation
Sigmoid function	$\phi(x) = \frac{1}{1 + e^{-x}}$	
Hyperbolic tangent function	$\phi(x) = \frac{2}{1 + e^{-2x}} - 1$	
Rectified linear unit (ReLU) function	$\phi(x) = \max(0, x)$	
Sign function	$\phi(x) = \text{sign}(x)$	